

DLT (Direct Linear Transform) method for estimating the camera projection matrix  $P$ .

The key idea is:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = P \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

where

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix}.$$

## 1. How many point correspondences?

There are 12 unknown  $p_{ij}$ :

$$p_{11}, p_{12}, \dots, p_{34}.$$

But  $P$  is a homogeneous matrix, so multiplying every element by the same constant gives the same camera. Therefore, there are effectively **11 independent unknowns**.

Each 3D–2D correspondence gives **2 independent equations**:

$$u = \frac{p_{11}X + p_{12}Y + p_{13}Z + p_{14}}{p_{31}X + p_{32}Y + p_{33}Z + p_{34}}$$

$$v = \frac{p_{21}X + p_{22}Y + p_{23}Z + p_{24}}{p_{31}X + p_{32}Y + p_{33}Z + p_{34}}.$$

So:

$$2N \geq 11$$

which means the theoretical minimum is

$$N = 6 \text{ point correspondences}$$

because 5 points give only 10 equations, while 6 give 12.

**In practice, use more than 6 points**—e.g. 10, 20, or more—spread throughout the 3D scene, and solve using SVD/least squares.

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## 2. Simple numerical example

Let's make the camera matrix extremely simple:

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

This is a simple camera where

$$u = \frac{X}{Z}, \quad v = \frac{Y}{Z}.$$

Suppose we have this 3D point:

$$X_w = \begin{bmatrix} 2 \\ 4 \\ 8 \\ 1 \end{bmatrix}.$$

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Project it into the image:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 8 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}.$$

Convert from homogeneous coordinates:

$$u = \frac{2}{8} = 0.25$$

$$v = \frac{4}{8} = 0.5.$$

So our correspondence is:

$$(X, Y, Z) = (2, 4, 8) \leftrightarrow (u, v) = (0.25, 0.5)$$

### 3. How this becomes the matrix $A$

From the two equations for one point are:

$$Xu?$$

More precisely:

$$Xp_{11} + Yp_{12} + Zp_{13} + p_{14} - u(Xp_{31} + Yp_{32} + Zp_{33} + p_{34}) = 0$$

and

$$Xp_{21} + Yp_{22} + Zp_{23} + p_{24} - v(Xp_{31} + Yp_{32} + Zp_{33} + p_{34}) = 0.$$

For our point

$$(X, Y, Z) = (2, 4, 8)$$

and

$$(u, v) = (0.25, 0.5),$$

the first equation becomes

$$2p_{11} + 4p_{12} + 8p_{13} + p_{14} - 0.25(2p_{31} + 4p_{32} + 8p_{33} + p_{34}) = 0.$$

Therefore our first row of  $A$  is

$$[2, 4, 8, 1, 0, 0, 0, 0, -0.5, -1, -2, -0.25]$$

The second equation gives

$$2p_{21} + 4p_{22} + 8p_{23} + p_{24} - 0.5(2p_{31} + 4p_{32} + 8p_{33} + p_{34}) = 0.$$

So the second row is

$$[0, 0, 0, 0, 2, 4, 8, 1, -1, -2, -4, -0.5]$$

Thus one correspondence produces two rows in  $A$ .

## 4. Six points

Suppose we have six correspondences:

| Point | 3D $(X, Y, Z)$ | Image $(u, v)$ |
|-------|----------------|----------------|
| 1     | (2, 4, 8)      | (0.25, 0.50)   |
| 2     | (4, 2, 8)      | (0.50, 0.25)   |
| 3     | (3, 6, 6)      | (0.50, 1.00)   |
| 4     | (1, 2, 4)      | (0.25, 0.50)   |
| 5     | (6, 3, 6)      | (1.00, 0.50)   |
| 6     | (2, 6, 4)      | (0.50, 1.50)   |

For example, point 2:

$$(X, Y, Z) = (4, 2, 8)$$

projects to

$$u = \frac{4}{8} = 0.5, \quad v = \frac{2}{8} = 0.25.$$

It generates these two rows:

$$[4, 2, 8, 1, 0, 0, 0, 0, -2, -1, -4, -0.5]$$

and

$$[0, 0, 0, 0, 4, 2, 8, 1, -1, -0.5, -2, -0.25].$$

Doing this for all six points gives

$$A_{12 \times 12} p = 0$$

where

$$p = [p_{11} \ p_{12} \ p_{13} \ p_{14} \ p_{21} \ p_{22} \ p_{23} \ p_{24} \ p_{31} \ p_{32} \ p_{33} \ p_{34}]^T.$$

Then you solve

$$Ap = 0$$

using SVD. The solution is the singular vector corresponding to the smallest singular value.

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## 5. The important takeaway

Think of it this way:

1 point  $\rightarrow$  2 equations

5 points  $\rightarrow$  10 equations

6 points  $\rightarrow$  12 equations

There are 12 entries in  $P$ , but because  $P$  is defined only up to scale, **6 correspondences is the minimum for DLT.**

### For a real calibration:

I'd recommend **at least 10–20 point correspondences**, and preferably more, with the 3D points well distributed in depth and across the image. This makes the SVD solution much more robust to measurement noise.

If you want, I can also take **6 actual numerical points and calculate the complete  $12 \times 12$   $A$  matrix and show how SVD recovers  $P$  step-by-step.**

Absolutely. Let's do a **complete numerical DLT example**, starting from 6 3D–2D correspondences, building the full  $12 \times 12$  matrix  $A$ , and then showing what SVD gives.

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## 1. Assume the actual camera matrix

For this example, let's choose a very simple camera:

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Therefore,

$$u = \frac{X}{Z}, \quad v = \frac{Y}{Z}.$$

We will pretend we don't know  $P$  and try to recover it from point correspondences.

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## 2. Give the camera 6 known 3D points

Let's use:

| Point | $x$ | $y$ | $z$ |
|-------|-----|-----|-----|
| 1     | 1   | 2   | 4   |
| 2     | 2   | 1   | 5   |
| 3     | 3   | 2   | 6   |
| 4     | 1   | 4   | 2   |
| 5     | 4   | 3   | 5   |
| 6     | 2   | 5   | 3   |

These are our **known world coordinates**.

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### 3. Calculate their image coordinates

Since

$$u = \frac{X}{Z}, \quad v = \frac{Y}{Z},$$

for point 1:

$$(X, Y, Z) = (1, 2, 4)$$

we get

$$u = \frac{1}{4} = 0.25$$

$$v = \frac{2}{4} = 0.5.$$

Do this for all points:

| Point | 3D point $(X, Y, Z)$ | $u$    | $v$    |
|-------|----------------------|--------|--------|
| 1     | (1, 2, 4)            | 0.25   | 0.50   |
| 2     | (2, 1, 5)            | 0.40   | 0.20   |
| 3     | (3, 2, 6)            | 0.50   | 0.3333 |
| 4     | (1, 4, 2)            | 0.50   | 2.00   |
| 5     | (4, 3, 5)            | 0.80   | 0.60   |
| 6     | (2, 5, 3)            | 0.6667 | 1.6667 |

Now we have our **six point correspondences**.

#### 4. Take point 1 and build two equations

Point 1 is:

$$(X, Y, Z) = (1, 2, 4)$$

and

$$(u, v) = (0.25, 0.5).$$

The DLT equations are:

$$Xp_{11} + Yp_{12} + Zp_{13} + p_{14} - u(Xp_{31} + Yp_{32} + Zp_{33} + p_{34}) = 0$$

and

$$Xp_{21} + Yp_{22} + Zp_{23} + p_{24} - v(Xp_{31} + Yp_{32} + Zp_{33} + p_{34}) = 0.$$

Substitute the numbers.

**Equation for  $u$**

$$p_{11} + 2p_{12} + 4p_{13} + p_{14} - 0.25p_{31} - 0.5p_{32} - p_{33} - 0.25p_{34} = 0$$

So the first row of  $A$  is:

$$[1, 2, 4, 1, 0, 0, 0, 0, -0.25, -0.5, -1, -0.25]$$

**Equation for  $v$**

$$p_{21} + 2p_{22} + 4p_{23} + p_{24} - 0.5p_{31} - p_{32} - 2p_{33} - 0.5p_{34} = 0$$

Therefore:

$$[0, 0, 0, 0, 1, 2, 4, 1, -0.5, -1, -2, -0.5]$$

So:

$$1 \text{ point} \rightarrow 2 \text{ rows of } A$$

## 5. Do this for all six points

We obtain the complete matrix:

$$A = \begin{bmatrix} 1 & 2 & 4 & 1 & 0 & 0 & 0 & 0 & -0.25 & -0.5 & -1 & -0.25 \\ 0 & 0 & 0 & 0 & 1 & 2 & 4 & 1 & -0.5 & -1 & -2 & -0.5 \\ 2 & 1 & 5 & 1 & 0 & 0 & 0 & 0 & -0.8 & -0.4 & -2 & -0.4 \\ 0 & 0 & 0 & 0 & 2 & 1 & 5 & 1 & -0.4 & -0.2 & -1 & -0.2 \\ 3 & 2 & 6 & 1 & 0 & 0 & 0 & 0 & -1.5 & -1 & -3 & -0.5 \\ 0 & 0 & 0 & 0 & 3 & 2 & 6 & 1 & -1 & -0.6667 & -2 & -0.3333 \\ 1 & 4 & 2 & 1 & 0 & 0 & 0 & 0 & -0.5 & -2 & -1 & -0.5 \\ 0 & 0 & 0 & 0 & 1 & 4 & 2 & 1 & -2 & -8 & -4 & -2 \\ 4 & 3 & 5 & 1 & 0 & 0 & 0 & 0 & -3.2 & -2.4 & -4 & -0.8 \\ 0 & 0 & 0 & 0 & 4 & 3 & 5 & 1 & -2.4 & -1.8 & -3 & -0.6 \\ 2 & 5 & 3 & 1 & 0 & 0 & 0 & 0 & -1.3333 & -3.3333 & -2 & -0.6667 \\ 0 & 0 & 0 & 0 & 2 & 5 & 3 & 1 & -1.3333 & -3.3333 & -2 & -1.6667 \end{bmatrix}$$

Notice its size:

$$A \text{ is } 12 \times 12$$

because:

$$6 \text{ points} \times 2 \text{ equations per point} = 12 \text{ equations.}$$

## 6. Now write $Ap = 0$

The unknown vector is

$$p = \begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \\ p_{14} \\ p_{21} \\ p_{22} \\ p_{23} \\ p_{24} \\ p_{31} \\ p_{32} \\ p_{33} \\ p_{34} \end{bmatrix}.$$

Therefore:

$$Ap = 0$$

or explicitly:

$$[12 \times 12][12 \times 1] = [12 \times 1].$$

## 7. But how do we solve $Ap = 0$ ?

This is where SVD comes in.

We calculate:

$$A = U\Sigma V^T.$$

We look at the singular value decomposition:

$$\Sigma = \begin{bmatrix} \sigma_1 & & & & \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_{11} & \\ & & & & \sigma_{12} \end{bmatrix}$$

and find the **smallest** singular value.

The corresponding column of  $V$  gives our solution  $p$ .

For this perfect, noise-free example, the result is proportional to:

$$p = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

Depending on how SVD normalizes it, you might actually see:

$$p \approx \begin{bmatrix} 0.57735 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0.57735 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0.57735 \\ 0 \end{bmatrix}.$$

The 0.57735 doesn't matter!

Why?

Because  $P$  is only defined **up to scale**.

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## 8. Reshape the answer back into $P$

Remember our vector was ordered as:

$$[p_{11}, p_{12}, p_{13}, p_{14}, p_{21}, p_{22}, p_{23}, p_{24}, p_{31}, p_{32}, p_{33}, p_{34}]^T.$$

So we reshape it:

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

🎉 We recovered the original camera matrix!

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## 9. You can verify it with one point

Take:

$$X = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 1 \end{bmatrix}.$$

Multiply:

$$PX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

Convert homogeneous coordinates:

$$u = \frac{1}{4} = 0.25$$

$$v = \frac{2}{4} = 0.5.$$

Exactly our original image point:

$$(u, v) = (0.25, 0.5)$$

## 10. The whole DLT procedure in one picture

You can think of the process as:

3D points + 2D points

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2 equations per correspondence

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$$Ap = 0$$

↓

$$A = U\Sigma V^T$$

↓

take  $V$ 's last column

↓

$$p$$

↓

reshape  $p \rightarrow P$

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